E1 Trigonometry — Worksheet

IGCSE Mathematics 0580 | Practice Problems

E1_Q1: Pythagoras Theorem - Problems

F Foundation [1 mark]

Q1. In a right-angled triangle, the two shorter sides are 3 cm and 4 cm. Find the length of the hypotenuse.

F Foundation [1 mark]

Q2. A right-angled triangle has a hypotenuse of 13 cm and one leg of 5 cm. Find the length of the other leg.

F Foundation [1 mark]

Q3. Find the length of the hypotenuse of a right-angled triangle with legs 6 cm and 8 cm.

F Foundation [1 mark]

Q4. A right-angled triangle has legs of lengths 9 cm and 12 cm. Determine the length of the hypotenuse.

C Core [1 mark]

Q5. A ladder of length 5 m rests against a vertical wall. The foot of the ladder is 1.4 m from the base of the wall. How far up the wall does the ladder reach? Give your answer correct to 1 decimal place.

C Core [1 mark]

Q6. A rectangular field is 80 m long and 60 m wide. A diagonal path runs from one corner to the opposite corner. Find the length of the path in metres.

C Core [1 mark]

Q7. A ship sails 24 km due east, then 10 km due north. How far is the ship from its starting point?

C Core [1 mark]

Q8. A rectangular box has dimensions 3 cm by 4 cm by 12 cm. Find the length of the longest diagonal from one corner to the opposite corner inside the box.

C Core [1 mark]

Q9. A right square pyramid has a square base of side 8 cm and a slant height (from apex to midpoint of base edge) of 10 cm. Find the height of the pyramid.

C Core [1 mark]

Q10. Two points A and B are 150 m apart on level ground. Point C is directly above point A at a height of 40 m. Point D is directly above point B at a height of 70 m. Find the straight-line distance between points C and D.

C Core [1 mark]

Q11. A room is 6 m long, 5 m wide, and 3 m high. Find the length of the longest rod that can fit inside the room.

C Core [1 mark]

Q12. A 50-foot wire is attached from the top of a 40-foot vertical pole to a stake in the ground. How far from the base of the pole is the stake?

H Challenge [1 mark]

Q13. A cuboid has sides of lengths 2x cm, 3x cm, and 6x cm. If the longest diagonal of the cuboid is 21 cm, find the value of x.

H Challenge [1 mark]

Q14. A right-angled triangle has a hypotenuse of (x + 6) cm and legs of (x + 1) cm and (x + 5) cm. Find the value of x.

H Challenge [1 mark]

Q15. From point P, a person walks 12 km due east to point Q, then 5 km due south to point R, and finally 8 km due west to point S. Find the straight-line distance from P to S.

H Challenge [1 mark]

Q16. A rectangular prism has length 8 cm, width 6 cm, and height h cm. If the space diagonal is double the height, find the value of h.

E1_Q2: Trigonometric Ratios - Problems

F Foundation [1 mark]

Q17. In a right-angled triangle, the sides are: opposite = 3, adjacent = 4, hypotenuse = 5. Find sin theta where theta is the angle opposite the side of length 3.

F Foundation [1 mark]

Q18. In a right-angled triangle with adjacent = 5 and hypotenuse = 13, find cos theta.

F Foundation [1 mark]

Q19. In a right-angled triangle, opposite = 8, adjacent = 15. Find tan theta.

F Foundation [1 mark]

Q20. A right-angled triangle has opposite = 7, hypotenuse = 25. Find sin theta.

C Core [1 mark]

Q21. In a right-angled triangle, sin theta = 0.5. Find the value of angle theta in degrees.

C Core [1 mark]

Q22. In triangle ABC, right-angled at B, AB = 8 cm and BC = 6 cm. Find angle ACB correct to 1 decimal place.

C Core [1 mark]

Q23. In a right-angled triangle, the hypotenuse is 10 cm and one angle is 40 degrees. Find the length of the side opposite this angle, correct to 2 decimal places.

C Core [1 mark]

Q24. In a right-angled triangle, the adjacent side to a 35-degree angle is 12 cm. Find the length of the hypotenuse, correct to 2 decimal places.

C Core [1 mark]

Q25. A right-angled triangle has hypotenuse of length 17 cm and one angle of 28 degrees. Find the length of the side adjacent to the 28-degree angle, correct to 2 decimal places.

C Core [1 mark]

Q26. In triangle PQR, right-angled at Q, PQ = 9 cm and PR = 15 cm. Find angle PRQ correct to 1 decimal place.

C Core [1 mark]

Q27. A ladder leaning against a wall makes an angle of 65 degrees with the ground. The ladder reaches 8 m up the wall. Find the length of the ladder, correct to 2 decimal places.

C Core [1 mark]

Q28. From a point on level ground, the angle of elevation to the top of a flagpole is 22 degrees. The point is 30 m from the base of the flagpole. Find the height of the flagpole, correct to 2 decimal places.

H Challenge [1 mark]

Q29. In a right-angled triangle, cos A = 3/5. Find the value of sin A and tan A without using a calculator.

H Challenge [1 mark]

Q30. In a right-angled triangle, 5 sin theta = 3. Find the exact values of cos theta and tan theta.

H Challenge [1 mark]

Q31. A right-angled triangle has area 54 cm^2. One acute angle is 36.87 degrees. Find the length of the hypotenuse, correct to 2 decimal places.

H Challenge [1 mark]

Q32. If tan theta = 3/4 and theta is an acute angle, find the exact values of sin theta and cos theta.

E1_Q3: Angle of Elevation and Depression - Problems

F Foundation [1 mark]

Q33. A person looks up at the top of a tree. The angle of elevation is 30 degrees and the person is 20 m from the base of the tree. Find the height of the tree, correct to 1 decimal place.

F Foundation [1 mark]

Q34. A flagpole casts a shadow 12 m long when the angle of elevation of the sun is 45 degrees. Find the height of the flagpole.

F Foundation [1 mark]

Q35. From the top of a cliff 50 m high, the angle of depression of a boat at sea is 30 degrees. Find the horizontal distance from the boat to the base of the cliff, correct to 1 decimal place.

F Foundation [1 mark]

Q36. A building is 30 m tall. From a point on the ground, the angle of elevation to the top of the building is 60 degrees. Find the distance from the point to the base of the building, correct to 1 decimal place.

C Core [1 mark]

Q37. From a point 25 m from the base of a building, the angle of elevation to the top of the building is 32 degrees. Find the height of the building, correct to 2 decimal places.

C Core [1 mark]

Q38. From the top of a 60 m lighthouse, the angle of depression of a buoy is 18 degrees. Find the horizontal distance from the buoy to the lighthouse, correct to 2 decimal places.

C Core [1 mark]

Q39. A kite is flying at a height of 40 m above the ground. The string is taut and makes an angle of 55 degrees with the horizontal. Find the length of the string, correct to 2 decimal places.

C Core [1 mark]

Q40. A surveyor measures the angle of elevation to the top of a tower as 27 degrees. She then moves 30 m closer to the tower and measures the angle of elevation as 42 degrees. Find the height of the tower, correct to 2 decimal places.

C Core [1 mark]

Q41. From the top of a vertical cliff 80 m high, the angles of depression of two boats on the same side of the cliff are 30 degrees and 45 degrees. Find the distance between the two boats, correct to 2 decimal places.

C Core [1 mark]

Q42. A boy flies a kite with 100 m of string. The angle of elevation of the kite is 38 degrees. Find the height of the kite above the ground, correct to 2 decimal places.

C Core [1 mark]

Q43. From the top of a 100 m building, the angle of depression to a car on the ground is 25 degrees. Find the distance from the car to the base of the building, correct to 2 decimal places.

C Core [1 mark]

Q44. A person stands at point A and measures the angle of elevation to the top of a 40 m tree as 28 degrees. How far is the person from the base of the tree, correct to 2 decimal places?

H Challenge [1 mark]

Q45. From a point A on level ground, the angle of elevation to the top of a tower is 18 degrees. From a point B, which is 50 m closer to the tower, the angle of elevation is 27 degrees. Find the height of the tower, correct to 2 decimal places.

H Challenge [1 mark]

Q46. From the top of a building 25 m tall, the angle of elevation to the top of a nearby tower is 15 degrees and the angle of depression to the base of the tower is 40 degrees. Find the height of the tower, correct to 2 decimal places.

H Challenge [1 mark]

Q47. Two points A and B are 80 m apart on level ground. The angles of elevation of a hot air balloon from A and B are 35 degrees and 52 degrees respectively. If the balloon is directly above the line AB, find the height of the balloon, correct to 2 decimal places.

H Challenge [1 mark]

Q48. From a point on the ground, the angle of elevation of a tower is 25 degrees. On walking 40 m towards the tower, the angle of elevation becomes 35 degrees. Find the height of the tower, correct to 2 decimal places.

E1_Q4: Bearings - Problems

F Foundation [1 mark]

Q49. A ship sails 5 km due north from port A to point B. What is the bearing of port A from point B?

F Foundation [1 mark]

Q50. Point P is 10 km due east of point Q. Find the bearing of P from Q.

F Foundation [1 mark]

Q51. A plane flies 8 km due west from airport X. What is the bearing of the plane from the airport?

F Foundation [1 mark]

Q52. The bearing of a ship from a lighthouse is 120 degrees. What is the bearing of the lighthouse from the ship?

C Core [1 mark]

Q53. A boat sails from port A on a bearing of 060 degrees for 20 km to point B. How far east of A is point B?

C Core [1 mark]

Q54. A ship sails 15 km on a bearing of 040 degrees. How far north has it travelled? Give your answer correct to 2 decimal places.

C Core [1 mark]

Q55. A plane flies 300 km on a bearing of 120 degrees. How far south of its starting point is it? Give your answer correct to 2 decimal places.

C Core [1 mark]

Q56. A ship sails from port P to Q on a bearing of 050 degrees for 30 km, then from Q to R on a bearing of 140 degrees for 40 km. Find the direct distance from P to R, correct to 2 decimal places.

C Core [1 mark]

Q57. A walker goes 8 km on a bearing of 030 degrees, then 6 km on a bearing of 120 degrees. Find the direct distance from the starting point, correct to 2 decimal places.

C Core [1 mark]

Q58. A helicopter flies 50 km on a bearing of 210 degrees. How far west and how far south has it travelled? Give both answers correct to 2 decimal places.

C Core [1 mark]

Q59. The bearing of a boat from a lighthouse is 075 degrees. What is the bearing of the lighthouse from the boat?

C Core [1 mark]

Q60. A ship sails from X to Y, 25 km on a bearing of 025 degrees. Find how far north Y is from X, correct to 2 decimal places.

H Challenge [1 mark]

Q61. A boat sails from harbour H to point A on a bearing of 060 degrees for 40 km, then from A to B on a bearing of 150 degrees for 30 km. Find the distance HB and the bearing of B from H.

H Challenge [1 mark]

Q62. A ship sails from port P to Q on a bearing of 045 degrees for 20 km, then from Q to R on a bearing of 135 degrees for 20 km. Find the distance PR and the bearing of R from P.

H Challenge [1 mark]

Q63. A ship sails from X to Y on a bearing of 030 degrees for 50 km, then from Y to Z on a bearing of 120 degrees for 40 km. Find the distance XZ, correct to 2 decimal places.

H Challenge [1 mark]

Q64. A ship sails 15 km from port A on a bearing of 060 degrees to point B, then 25 km on a bearing of 150 degrees to point C. Calculate the distance AC and the bearing of C from A, both correct to 2 decimal places.

E1_Q5: Sine Rule - Problems

F Foundation [1 mark]

Q65. In triangle ABC, angle A = 30 degrees, angle B = 45 degrees, and side a (opposite A) = 10 cm. Find side b (opposite B) using the sine rule. Give your answer correct to 2 decimal places.

F Foundation [1 mark]

Q66. In triangle PQR, angle P = 60 degrees, angle Q = 70 degrees, and side p (opposite P) = 12 cm. Find side q (opposite Q). Give your answer correct to 2 decimal places.

F Foundation [1 mark]

Q67. In triangle XYZ, angle X = 40 degrees, angle Y = 80 degrees, and side x (opposite X) = 8 cm. Find side y (opposite Y). Give your answer correct to 2 decimal places.

F Foundation [1 mark]

Q68. In triangle ABC, angle A = 35 degrees, side a = 7 cm, and side b = 10 cm. Find angle B. Give your answer correct to 1 decimal place.

C Core [1 mark]

Q69. In triangle ABC, side a = 8 cm, side b = 10 cm, and angle A = 30 degrees. Find angle B. Give your answer correct to 1 decimal place.

C Core [1 mark]

Q70. In triangle PQR, side p = 15 cm, side q = 20 cm, and angle P = 42 degrees. Find angle Q. Give your answer correct to 1 decimal place.

C Core [1 mark]

Q71. In triangle ABC, angle A = 50 degrees, angle B = 60 degrees, and side c = 12 cm. Find side a (opposite A). Give your answer correct to 2 decimal places.

C Core [1 mark]

Q72. In triangle XYZ, angle X = 65 degrees, angle Z = 45 degrees, and side y = 20 cm. Find side x (opposite X). Give your answer correct to 2 decimal places.

C Core [1 mark]

Q73. In triangle ABC, side a = 9 cm, side b = 12 cm, and angle A = 25 degrees. Find angle B, correct to 1 decimal place.

C Core [1 mark]

Q74. In triangle ABC, side a = 14 cm, angle B = 55 degrees, angle C = 65 degrees. Find side b, correct to 2 decimal places.

C Core [1 mark]

Q75. In triangle PQR, angle Q = 72 degrees, angle R = 48 degrees, and side p = 25 cm. Find side q, correct to 2 decimal places.

C Core [1 mark]

Q76. In triangle XYZ, side x = 6 cm, side y = 8 cm, and angle X = 28 degrees. Find angle Y, correct to 1 decimal place.

H Challenge [1 mark]

Q77. In triangle ABC, side a = 10 cm, side b = 15 cm, and angle A = 30 degrees. Find the possible values of angle B, correct to 1 decimal place.

H Challenge [1 mark]

Q78. In triangle ABC, angle A = 40 degrees, angle B = 65 degrees, and side a = 30 cm. Find the length of the longest side, correct to 2 decimal places.

H Challenge [1 mark]

Q79. A surveyor needs to find the distance across a lake. She measures a baseline of 200 m along the shore. From one end of the baseline, the angle to a tree across the lake is 72 degrees. From the other end, the angle to the same tree is 48 degrees. Find the distance from the nearer end of the baseline to the tree, correct to 2 decimal places.

H Challenge [1 mark]

Q80. In triangle ABC, angle A = 36 degrees, side a = 8 cm, and side b = 12 cm. How many triangles are possible? Find all possible values of angle B, correct to 1 decimal place.

E1_Q6: Cosine Rule — Problems

F Foundation [3 marks]

Q81. In triangle ABC, AB = 6 cm, AC = 8 cm and angle BAC = 60°. Find BC using the cosine rule.

F Foundation [3 marks]

Q82. In triangle PQR, PQ = 5 cm, PR = 7 cm and QR = 8 cm. Find the angle at P.

F Foundation [3 marks]

Q83. A triangle has sides of length 4 cm, 5 cm and 6 cm. Use the cosine rule to find the largest angle.

F Foundation [3 marks]

Q84. Find the side length a in a triangle where b = 9 cm, c = 7 cm and angle A = 48°.

C Core [4 marks]

Q85. A ship sails 12 km east, then 15 km on a bearing of 060°. How far is it from its starting point?

C Core [5 marks]

Q86. Triangle ABC has AB = 11 cm, AC = 9 cm and angle BAC = 72°. Find BC and the angle at B.

C Core [4 marks]

Q87. An isosceles triangle has equal sides 10 cm and base 12 cm. Find the apex angle.

C Core [4 marks]

Q88. A garden triangular lawn has sides 8 m, 11 m and 14 m. Find the smallest angle of the triangle.

H Challenge [6 marks]

Q89. In triangle ABC, AB = 8 cm, AC = 11 cm and angle B = 38°. Find all possible values of side BC.

H Challenge [6 marks]

Q90. A quadrilateral ABCD has AB = 5 cm, BC = 7 cm, CD = 9 cm, DA = 6 cm and diagonal AC = 8 cm. Find angle BCD.

H Challenge [6 marks]

Q91. A right square-based pyramid has base side 6 cm and slant height 10 cm. Find the angle between a slant edge and the base plane.

H Challenge [5 marks]

Q92. A plane flies 200 km on bearing 050°, then 150 km on bearing 140°. Find its distance and bearing from the starting point.

X Extension [5 marks]

Q93. Prove the cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$ for any triangle ABC.

X Extension [5 marks]

Q94. A triangle has two sides of length 8 cm and 12 cm. The included angle varies. Find the maximum possible length of the third side.

X Extension [6 marks]

Q95. In quadrilateral ABCD, AB = 6 cm, BC = 8 cm, CD = 10 cm, DA = 7 cm and diagonal BD = 9 cm. Find angle ABC.

X Extension [5 marks]

Q96. A triangle has sides of length a, b and a+b. Show that the angle opposite the longest side must be greater than 120°.

E1_Q7: Area of Triangle using 1/2 ab sin C - Problems

F Foundation [1 mark]

Q97. A triangle has two sides of lengths 6 cm and 8 cm with an included angle of 30 degrees. Find the area of the triangle.

F Foundation [1 mark]

Q98. Calculate the area of a triangle with sides 10 cm and 12 cm and an included angle of 45 degrees. Give your answer correct to 2 decimal places.

F Foundation [1 mark]

Q99. Find the area of a triangle with sides 5 cm and 9 cm and an included angle of 60 degrees.

F Foundation [1 mark]

Q100. A triangle has sides of 7 cm and 11 cm with an included angle of 90 degrees. Find its area.

C Core [1 mark]

Q101. A triangle has two sides of lengths 8 cm and 12 cm with an included angle of 50 degrees. Find the area, correct to 2 decimal places.

C Core [1 mark]

Q102. A triangle has area 30 sq cm and two sides of lengths 8 cm and 10 cm. Find the included angle between these sides, correct to 1 decimal place.

C Core [1 mark]

Q103. A triangle has area 25 sq cm, one side of length 5 cm, and an included angle of 30 degrees. Find the length of the other side that forms the angle.

C Core [1 mark]

Q104. Find the area of a triangle with sides 9 cm and 14 cm and an included angle of 65 degrees. Give your answer correct to 2 decimal places.

C Core [1 mark]

Q105. A triangle has area 40 sq cm and two sides of lengths 10 cm and 12 cm. Find the possible values of the included angle between these sides, correct to 1 decimal place.

C Core [1 mark]

Q106. Calculate the area of a triangle with sides 6 cm and 10 cm and an included angle of 40 degrees. Give your answer correct to 2 decimal places.

C Core [1 mark]

Q107. A triangle has area 36 sq cm, one side of length 9 cm, and the included angle between this side and the unknown side is 55 degrees. Find the length of the unknown side, correct to 2 decimal places.

C Core [1 mark]

Q108. Find the area of a triangle with sides 15 cm and 20 cm and an included angle of 35 degrees. Give your answer correct to 2 decimal places.

H Challenge [1 mark]

Q109. In triangle ABC, side a = 12 cm, side b = 15 cm, and side c = 10 cm. Find the area of the triangle using the formula (1/2)ab sin C. First find angle C using the cosine rule. Give your answer correct to 2 decimal places.

H Challenge [1 mark]

Q110. In triangle ABC, angle A = 40 degrees, side b = 20 cm, and side c = 25 cm. Find the area of the triangle, correct to 2 decimal places.

H Challenge [1 mark]

Q111. In triangle PQR, side p = 14 cm, side q = 18 cm, and the area is 100 sq cm. Find the included angle R between sides p and q, correct to 1 decimal place.

H Challenge [1 mark]

Q112. In triangle ABC, sides a = 13 cm, b = 14 cm, and c = 15 cm. Find the area of the triangle. (Hint: Use cosine rule to find one angle, then use area formula.) Give your answer correct to 2 decimal places.